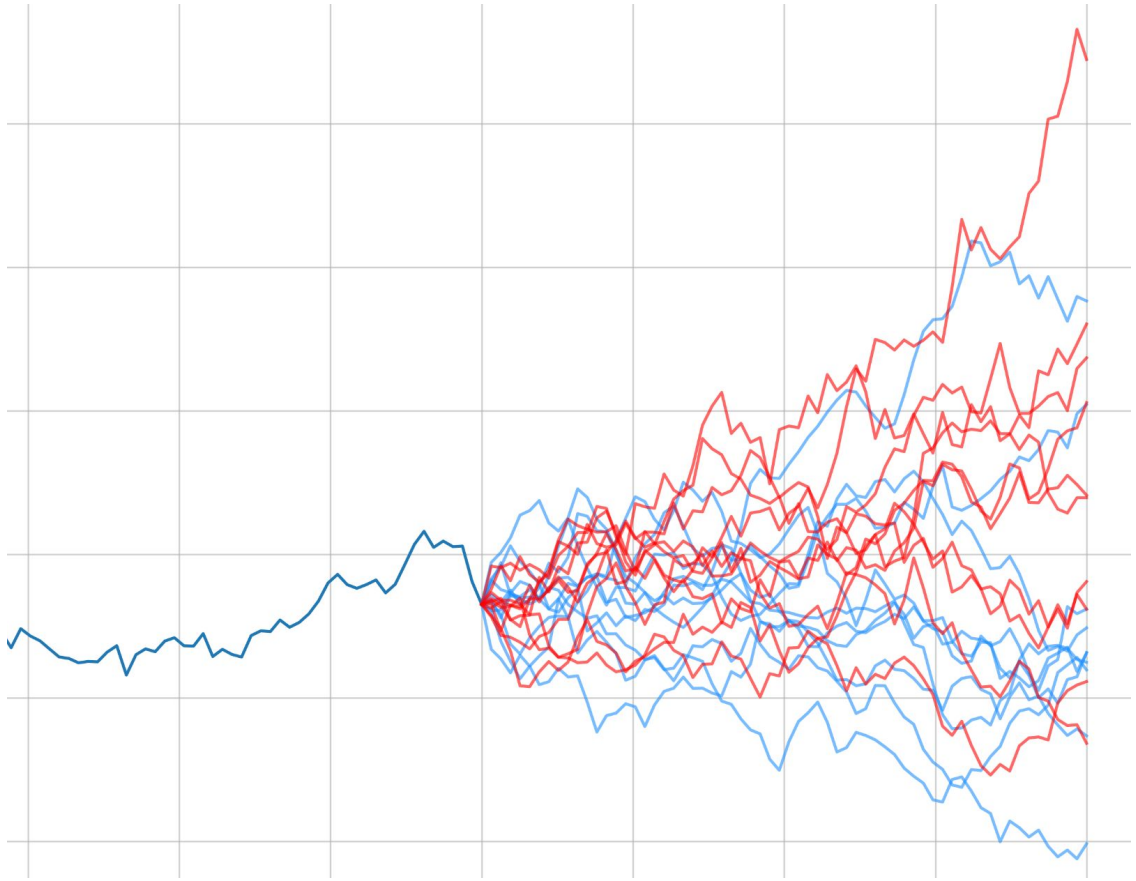
Continuous-Time Generative Models of Stochastic Processes based on Neural SDE



Maksim KALEDIN, Faculty of Computer Science, HSE University

Stochastics, Artificial Intelligence and Mathematical Finance 2026

Neural SDE Team



Anastasia BELOTSERKOVETS
Stochastics Expert

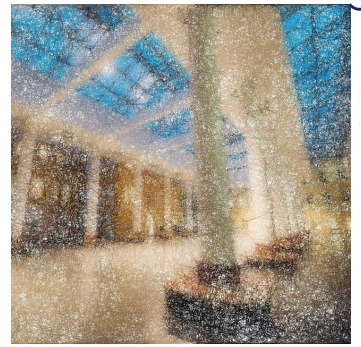


Nikita CHUIKIN
Stochastics Expert



Maksim KALEDIN
Crazy pipeline programming and problem stating

+ Neural (Jump)SDE
+ Applications to Math Finance
and more…



Conditional Forecasts

Conditional forecasting: given history, sample from conditional distribution of the future

1. Monte-Carlo Control and Optimal stopping
2. Forecasting with Uncertainty Quantification

Goal: Nice simulation model with right conditional generation

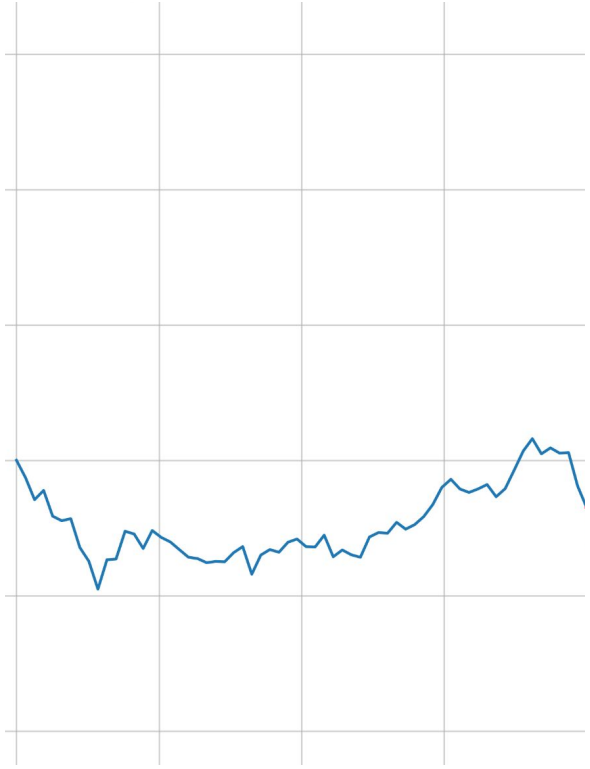


Conditional Forecasts

Conditional forecasting: given history, sample from conditional distribution of the future

1. Monte-Carlo Control and Optimal stopping
2. Forecasting with Uncertainty Quantification

Goal: Nice simulation model with right conditional generation



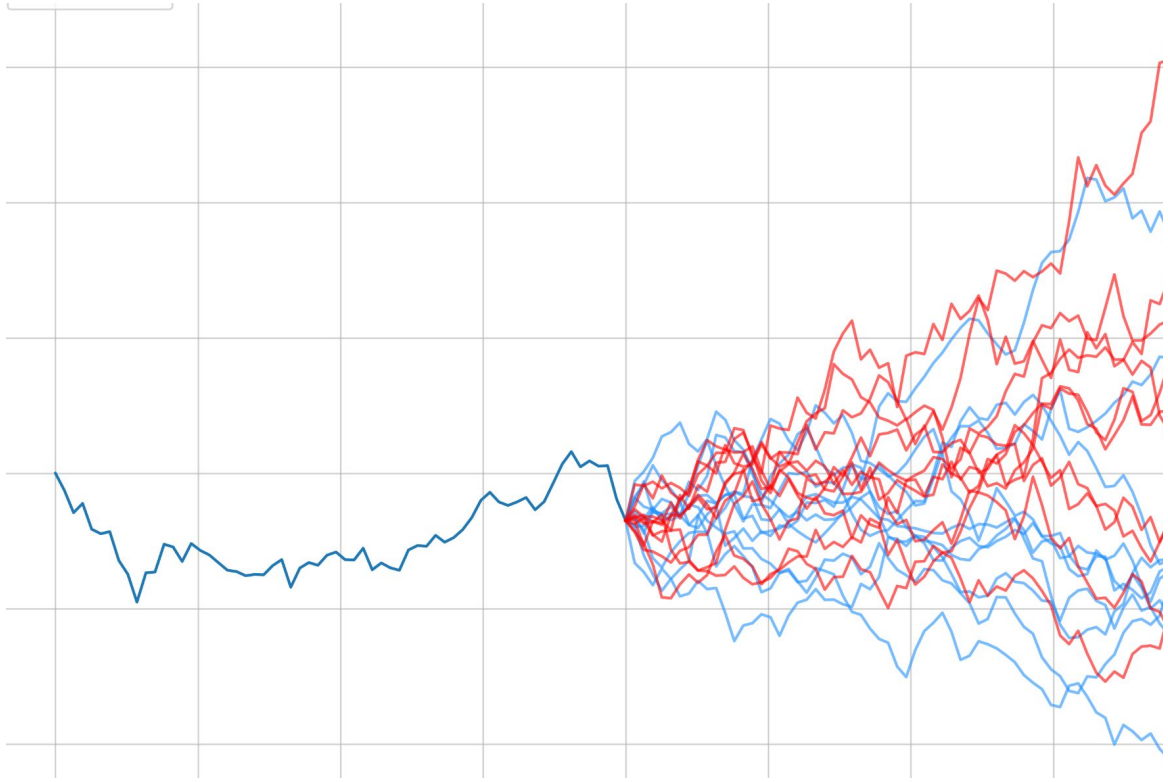


Conditional Forecasts

Conditional forecasting: given history, sample from conditional distribution of the future

1. Monte-Carlo Control and Optimal stopping
2. Forecasting with Uncertainty Quantification

Goal: Nice simulation model with right conditional generation

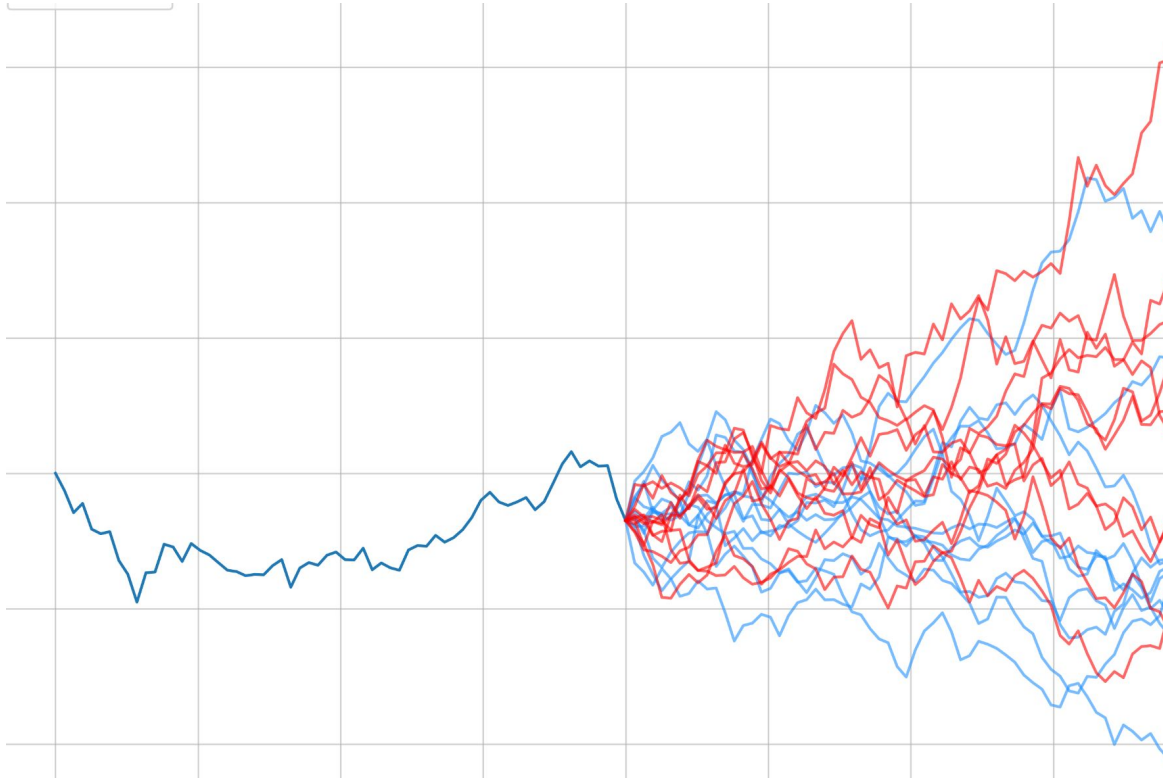


Conditional Forecasts

Conditional forecasting: given history, sample from conditional distribution of the future

1. Monte-Carlo Control and Optimal stopping
2. Forecasting with Uncertainty Quantification

Goal: Nice simulation model with right conditional generation



Formally: minimize the "distance" between finite-dimensional distributions

Neural ODEs

ODE:

time derivative

$$\dot{X} = f(t, X_t), \;\; X_0 = x_0$$



Neural ODEs

ODE:

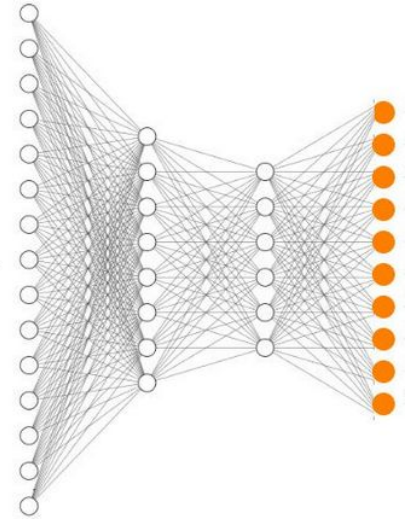
time derivative

$$\dot{X} = f(t, X_t), \;\; X_0 = x_0$$

Neural ODE:

$$\dot{X} = f_\theta(t, X_t), \;\; X_0 = x_0, \;\; \theta \in \mathbb{R}^N$$

Your fancy neural network



You All Have Seen “Neural” ODE

$$\dot{X} = f_\theta(t, X_t),\ X_0 = x_0,\ \theta \in \mathbb{R}^N$$

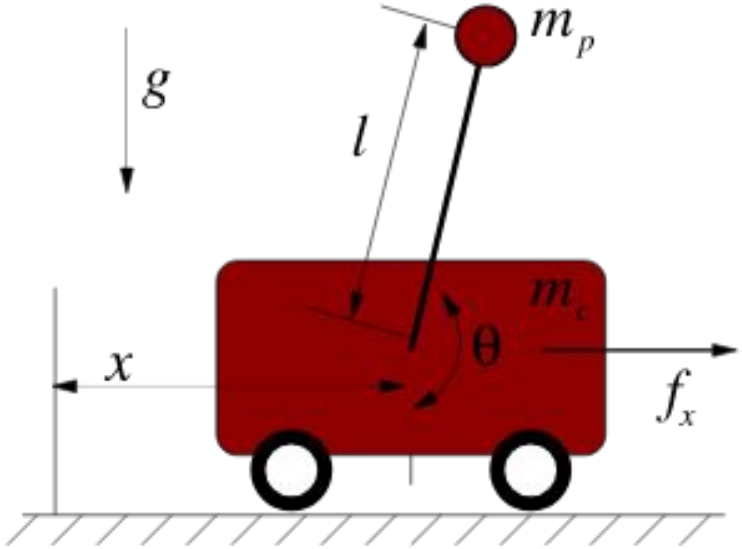


You All Have Seen “Neural” ODE

$$\dot{X} = f_\theta(t, X_t),\ X_0 = x_0,\ \theta \in \mathbb{R}^N$$

Example: Cart-Pole system

- Deterministic
- Parameters: mass, gravity, …
- Exists in continuous time



Neural ODEs

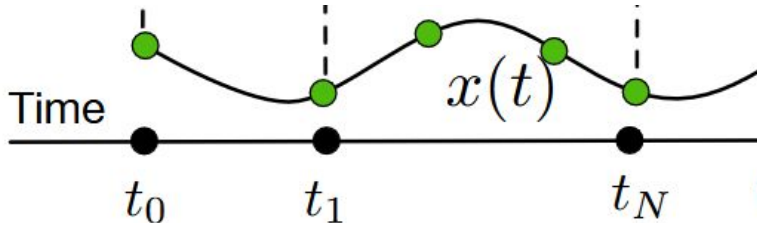
$$\dot{X} = f_\theta(t, X_t), \;\; X_0 = x_0, \;\; \theta \in \mathbb{R}^N$$

What for?..

1. **ODE** parameters hard to estimate (still people try) but produce smooth trajectories.

2. **Neural Networks** are powerful to model **X** but reality can be much smoother.

In the same time time series **data is discrete** because it is **collected in a discrete way**.



Neural ODEs

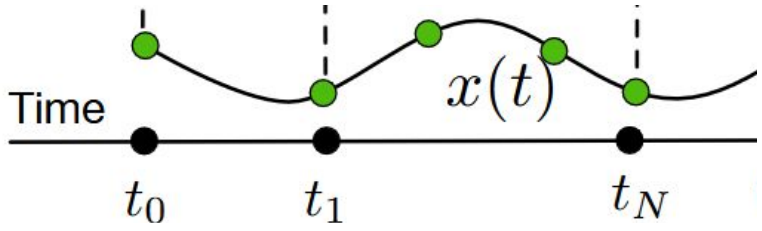
$$\dot{X} = f_\theta(t, X_t), \;\; X_0 = x_0, \;\; \theta \in \mathbb{R}^N$$

What for?..

1. **ODE** parameters hard to estimate (still people try) but produce smooth trajectories.

2. **Neural Networks** are powerful to model **X** but reality can be much smoother.

Combine **both** to get smooth trajectories and generality.



Neural ODE Training: First Attempts

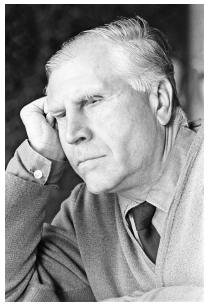
$$\dot{X} = f_\theta(t, X_t)$$

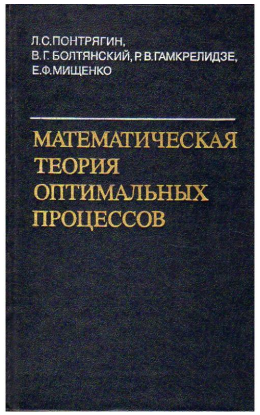


$$\begin{bmatrix} Y_n \\ H_{n+1} \end{bmatrix} = h(X_n, H_n)$$

Neural ODE Training: First Attempts

$$\dot{X} = f_\theta(t, X_t)$$





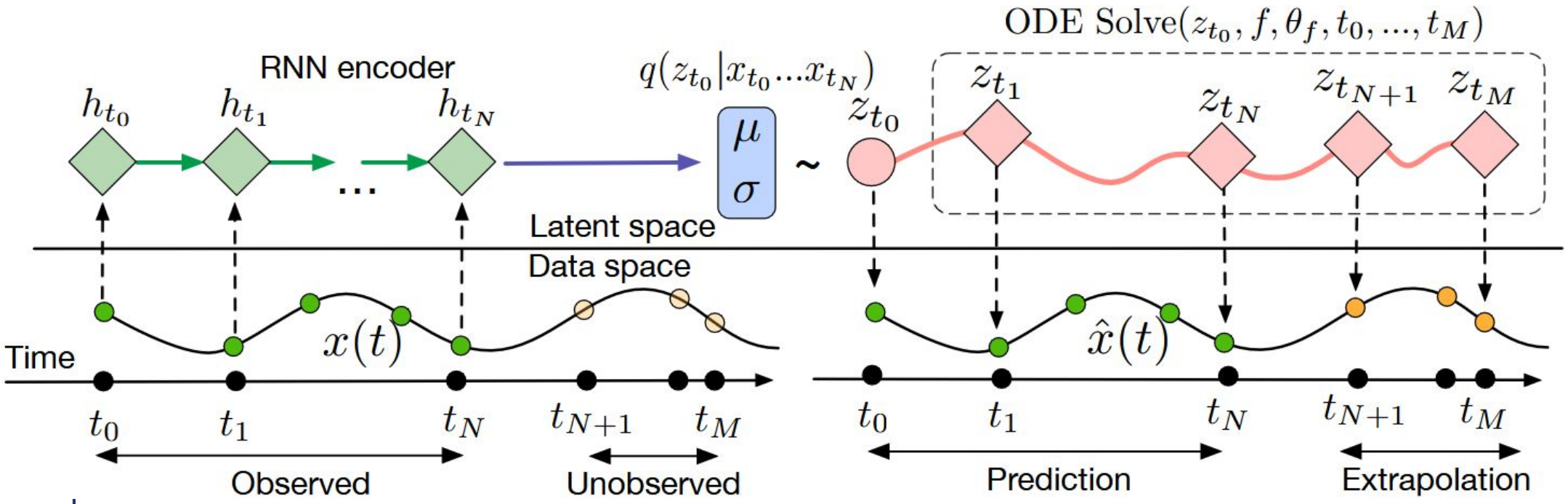
Chen et al. 2018



$$\begin{bmatrix} Y_n \\ H_{n+1} \end{bmatrix} = h(X_n, H_n)$$

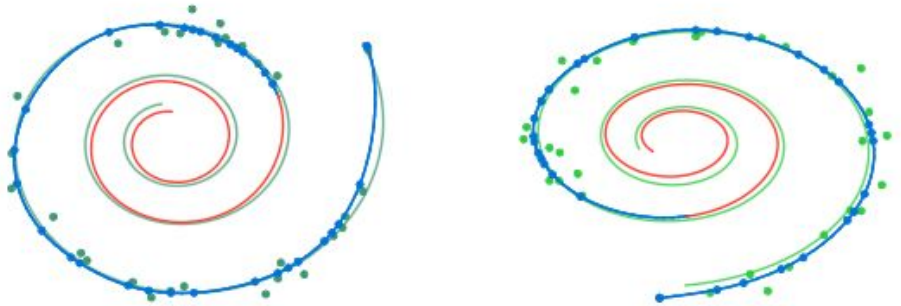
NeuralODE (Chen et al. 2018): Time Series Model

$$\mathbf{z}_{t_0} \sim p(\mathbf{z}_{t_0})$$
$$\mathbf{z}_{t_1}, \mathbf{z}_{t_2}, \ldots, \mathbf{z}_{t_N} = \text{ODESolve}(\mathbf{z}_{t_0}, f, \theta_f, t_0, \ldots, t_N)$$
$$\text{each} \quad \mathbf{x}_{t_i} \sim p(\mathbf{x}|\mathbf{z}_{t_i}, \theta_{\mathbf{x}})$$



Neural ODE Limitations

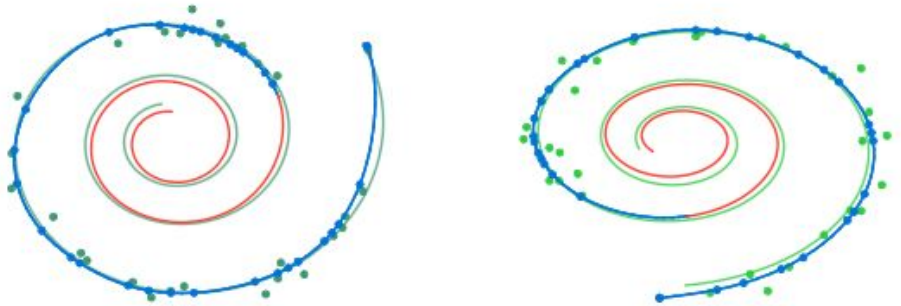
Limited modelling of stochasticity



Neural ODE Limitations

Limited modelling of stochasticity

In the meantime, Exchange:



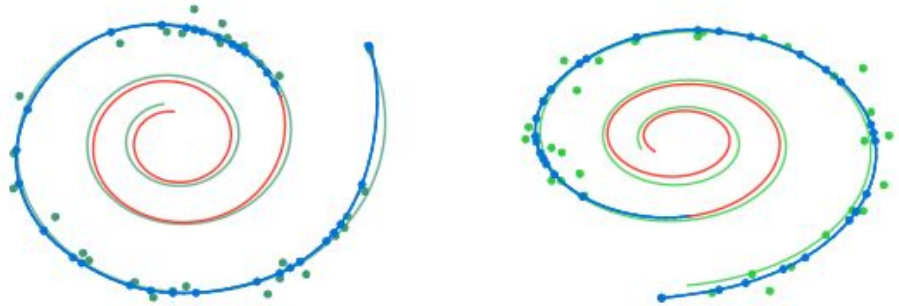


Neural ODE Limitations

Limited modelling of stochasticity

In the meantime, Exchange:





1. Volatility
2. Discontinuities
3. Highly Non-Stationary

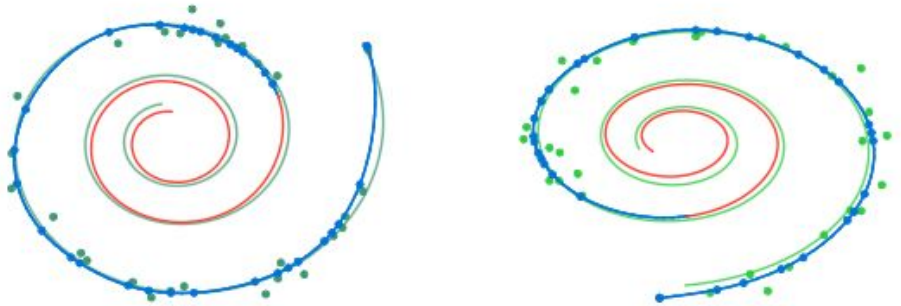
We need stochastic models to solve control, risk estimations, pricing etc

Neural ODE Limitations

Limited modelling of stochasticity

In the meantime, France Electricity:





1. Volatility
2. Discontinuities
3. Highly Non-Stationary

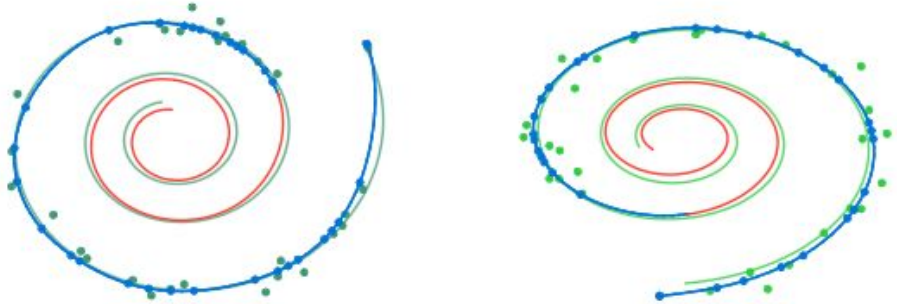
We need stochastic models to solve control, risk estimations, pricing etc

Neural ODE Limitations

Limited modelling of stochasticity

In the meantime, France Electricity:





1. Volatility
2. Discontinuities
3. Highly Non-Stationary

We need stochastic models to solve control, risk estimations, pricing etc

...conditional models (forecasting or interpolation)

Monte Carlo stuff

Neural SDEs

Your fancy neural network

Your fancy neural network

$$dX_t = \mu_\theta(t, X_t)dt + \sigma_\theta(t, X_t)dW_t, \quad X_0 = x_0$$

Wiener process

Just written like that, in fact it is

$$X_t = X_0 + \int_0^t \mu_\theta(s, X_s)ds + \int_0^t \sigma_\theta(s, X_s)dW_s$$

Ito or Stratonovich sense (we will assume Stratonovich).

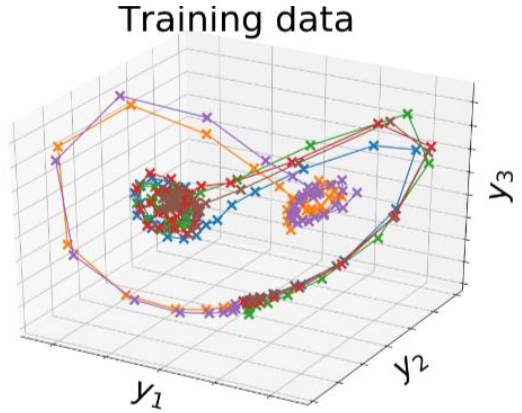


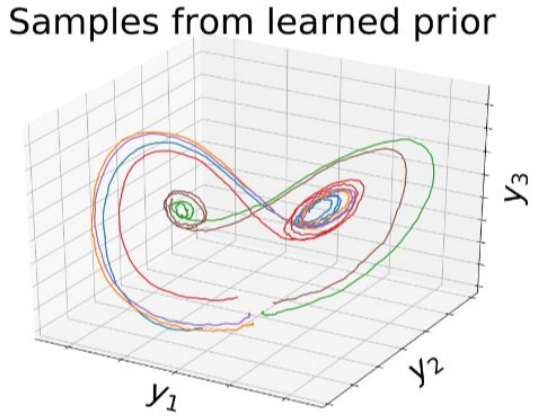
Neural SDE So Far

$$dX_t = \mu_\theta(t, X_t)dt + \sigma_\theta(t, X_t)dW_t, \quad X_0 = x_0$$

How to fit the model to the data?

One could discretize and proceed with usual sequence machine learning, but...





Neural SDE So Far

$$dX_t = \mu_\theta(t, X_t)dt + \sigma_\theta(t, X_t)dW_t, \quad X_0 = x_0$$

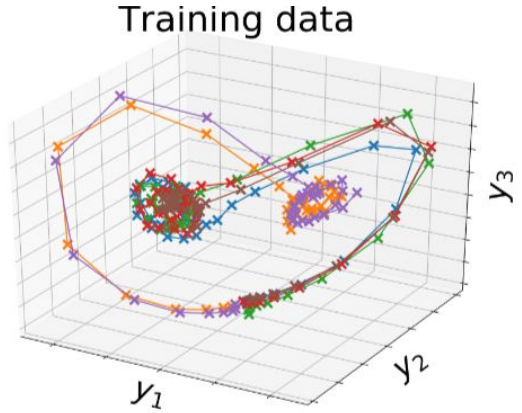
How to fit the model to the data?

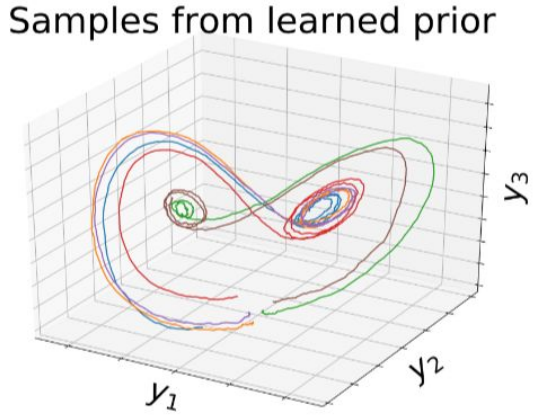
One could discretize and proceed with usual sequence machine learning, but…

Main Problems:

1. When using naive approaches, estimates are unstable or stochasticity tends to zero
2. Good predictions quality does not necessarily means good simulation model needed for downstream tasks.

Generative methodology is needed





Neural SDE So Far

$$dX_t = \mu_\theta(t, X_t)dt + \sigma_\theta(t, X_t)dW_t, \quad X_0 = x_0$$

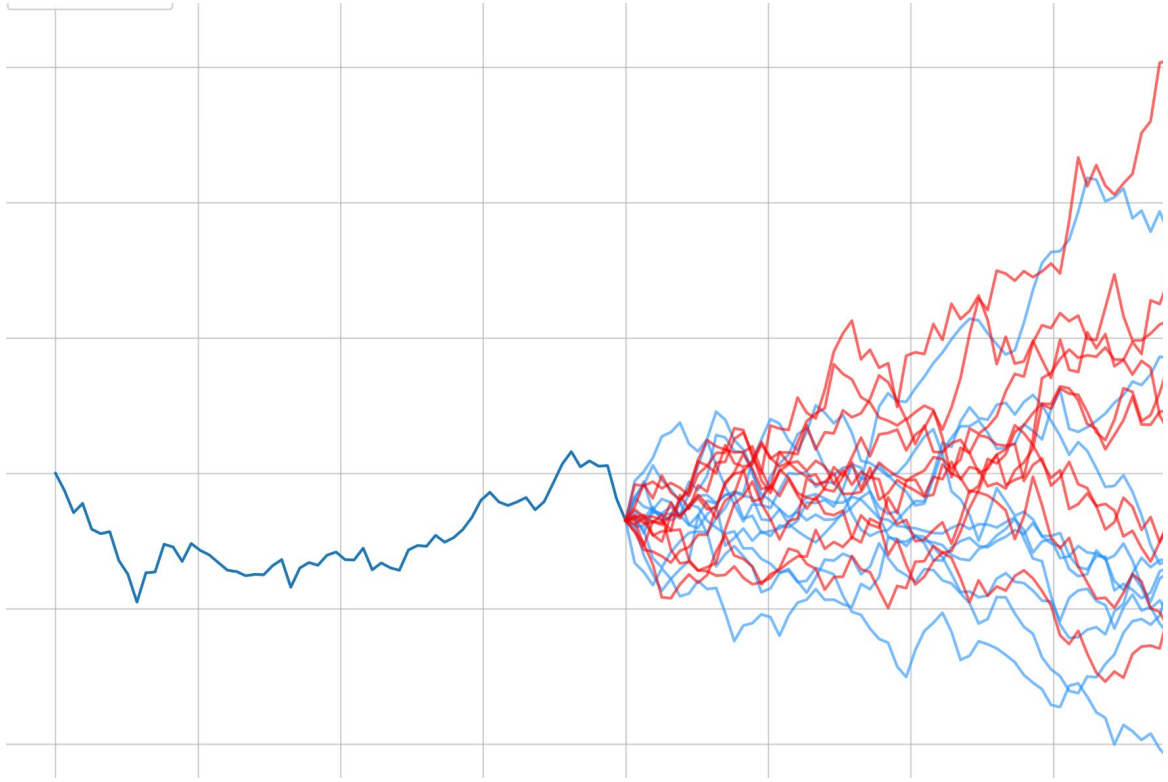
1. **Neural ODE** (Chen et al. 2018) and various extensions
2. **Latent SDE:** Fast simulation and backprop methods done for neural SDE (Li et al. 2020)
3. **SDE-GAN:** GAN-type generative model (Kidger et al. 2022)
4. Time series imputation and time series classification (Oh et al., 2024)
5. Simple video and pose interpolation (Bartosh et al. 2025)
6. **SDE Matching:** Optimizing computations in Latent SDE (Bartosh et al. 2025)
7. **Neural MJD:** Merton-type neural jump diffusion (Gao et al. 2025)

Conditional Forecasts

Conditional forecasting: given history, sample from conditional distribution of the future

1. Monte-Carlo Control and Optimal stopping
2. Forecasting with Uncertainty Quantification

Goal: Nice simulation model with right conditional generation



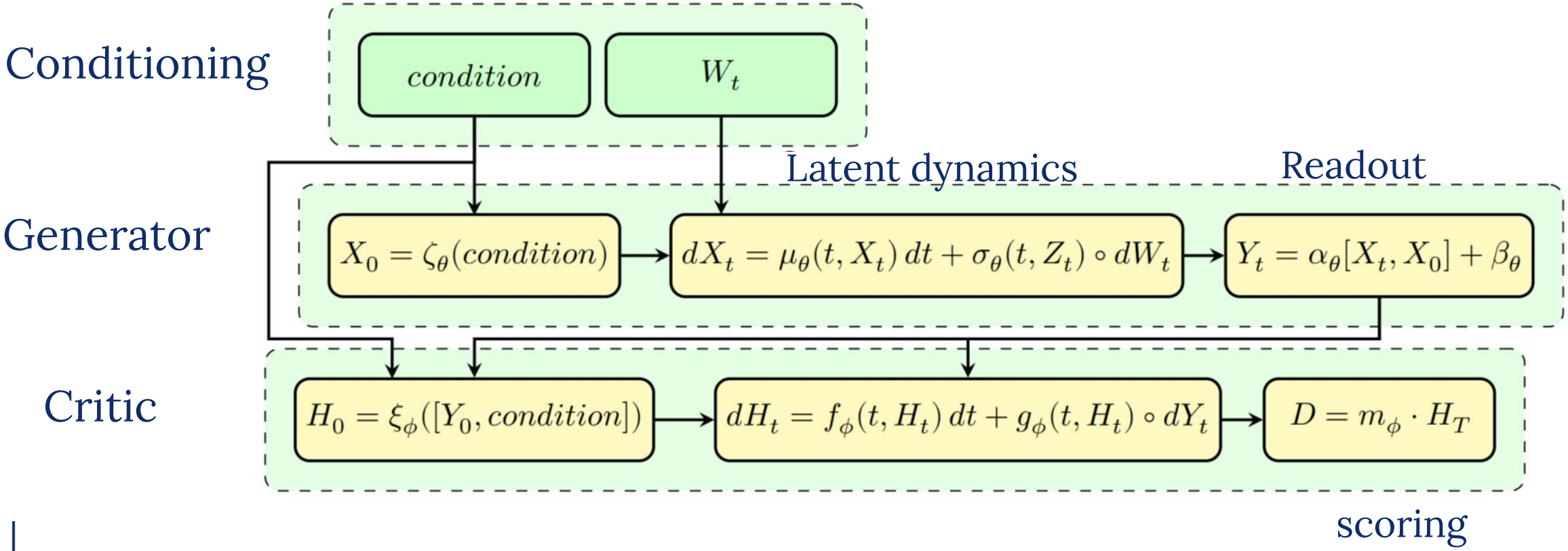
Formally: minimize the "distance" between finite-dimensional distributions

Conditional SDE-GAN (2026?)

Metrics.

1. Prediction quality: regression metrics (e.g. MAE)
2. Simulation quality: W1 distance and KS-statistics

Model: SDE-GAN with conditioning



Conditional SDE-GAN (2026?)

Training: Wasserstein GAN

$$W_1(\mu^C_{true}, \mu^C_{gen}) = \sup_{f \in Lip(1)} \mathbb{E}[f(X_{true}, C)] - \mathbb{E}[f(X_{gen}, C)]$$

Kantorovich-Rubinstein representation for W1 metric between distributions.

Main idea: enforce Lipschitz constraint on critic and run optimization in two time scales like in conventional GAN

- critic **(f)** maximizes the difference and approximates the W1 metric.
- generator uses it to minimize the Kantorovich-Rubinstein distance



Real Data Scenario

In reality there are usually no “many” trajectories, **just one** realization.

Most models focus on either

1. **Artificial examples**: learn model from bunch of trajectories

Or

2. **Not-quite-time-series data:** say, video with fixed set of objects

Or

3. **Time series, but produced in a way of independent sampling** (present in time series classification and anomaly detection).

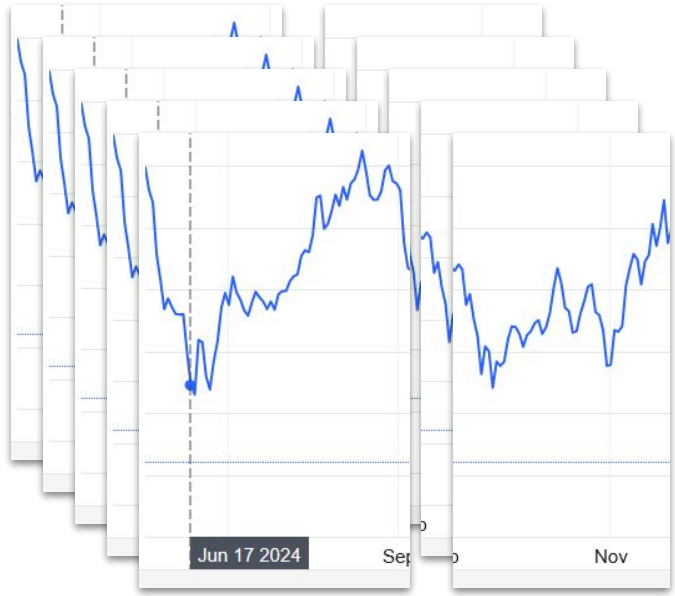


Real Data Scenario

In reality there are usually no “many” trajectories, **just one** realization.

Our proposal: use just one realization by cutting it into frames.

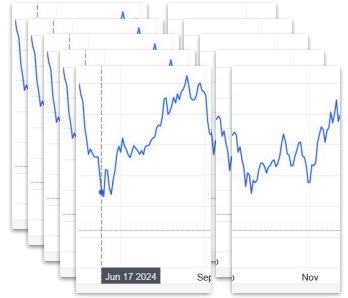




How reliable can this approach be?

Experiments

Generate one trajectory, separate it into train, val and test region in time.



Data:

- **GBM 1D**: t=[0,5] for training, t=[5,12] for testing, dt=5e-4
- **CIR 1D:** t=[0,5] for training, t=[5,12] for testing, dt=5e-3

~1e3..1e4 observations for training

Additionally subsamples are generated for every window.

Models:

1. **Trivial baseline:** VAR(1)
2. **Oracle baseline:** MLE
3. **Our model:** Conditional SDE GAN (~1 layer MLP for drift and diffusion with 16d latent space)

Metrics: MAE (prediction quality), W1 (distribution quality)

Experiments (Tables)

Method	MAE(avg)	W1(avg)
MLE	91.389	34.282
AR(1)	173.023	171.243
SDE-GAN	91.434	35.342

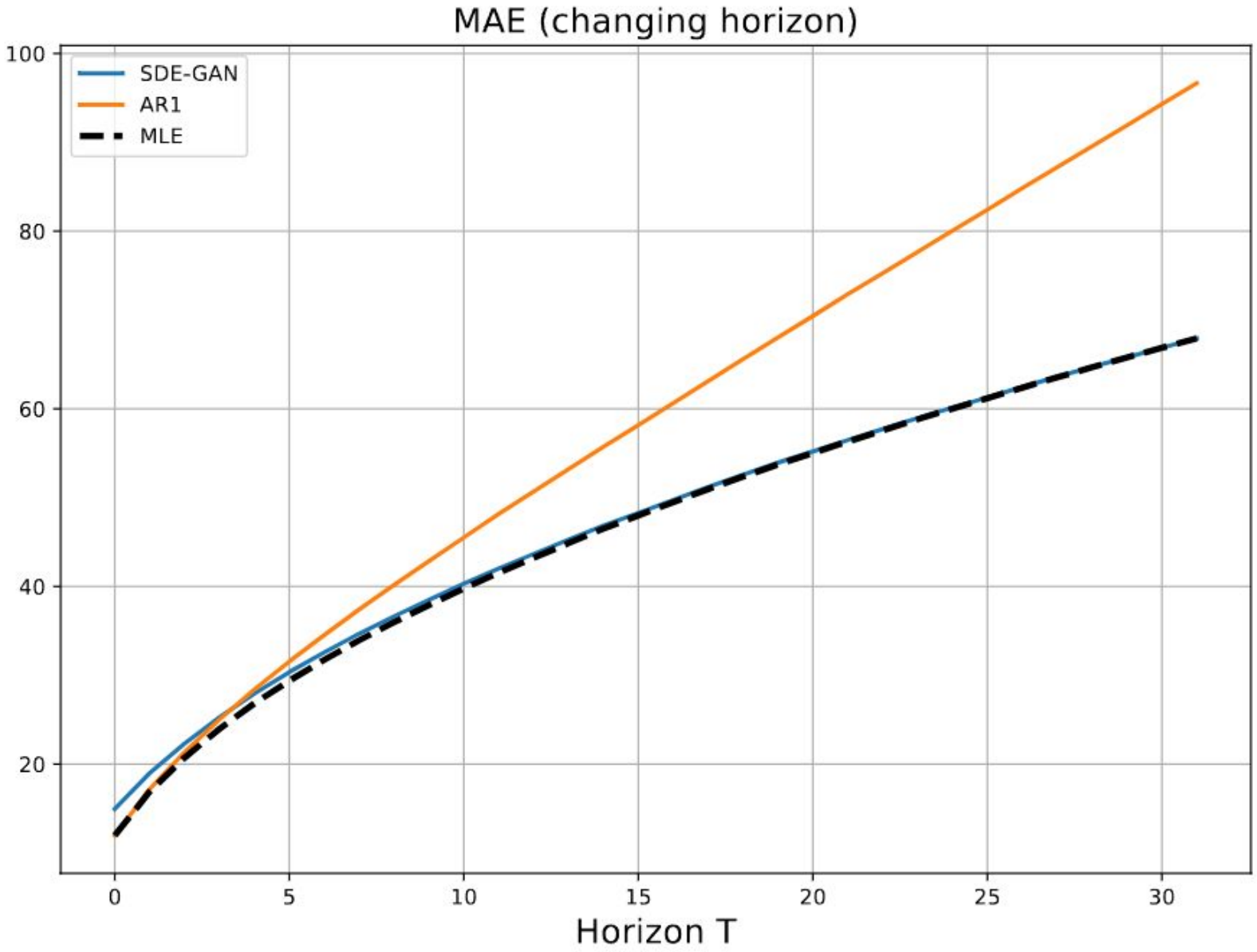
GBM (1D, test, prediction=128 steps)

Method	MAE(avg)	W1(avg)
MLE (!!)	0.659	0.214
AR(1)	15.237	15.896
SDE-GAN	0.649	0.185

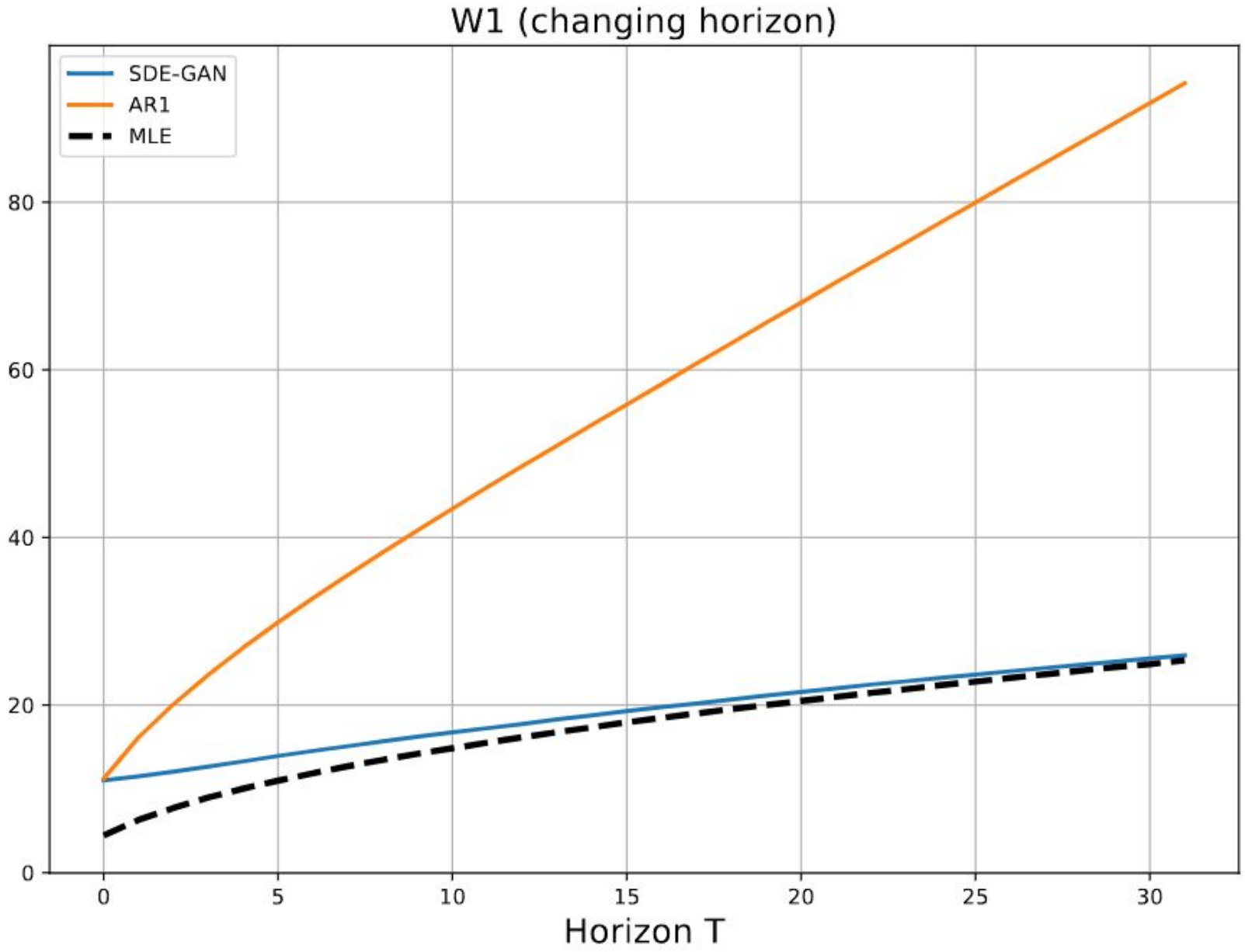
CIR (1D, test, prediction=256 steps)
Best MLE try is taken



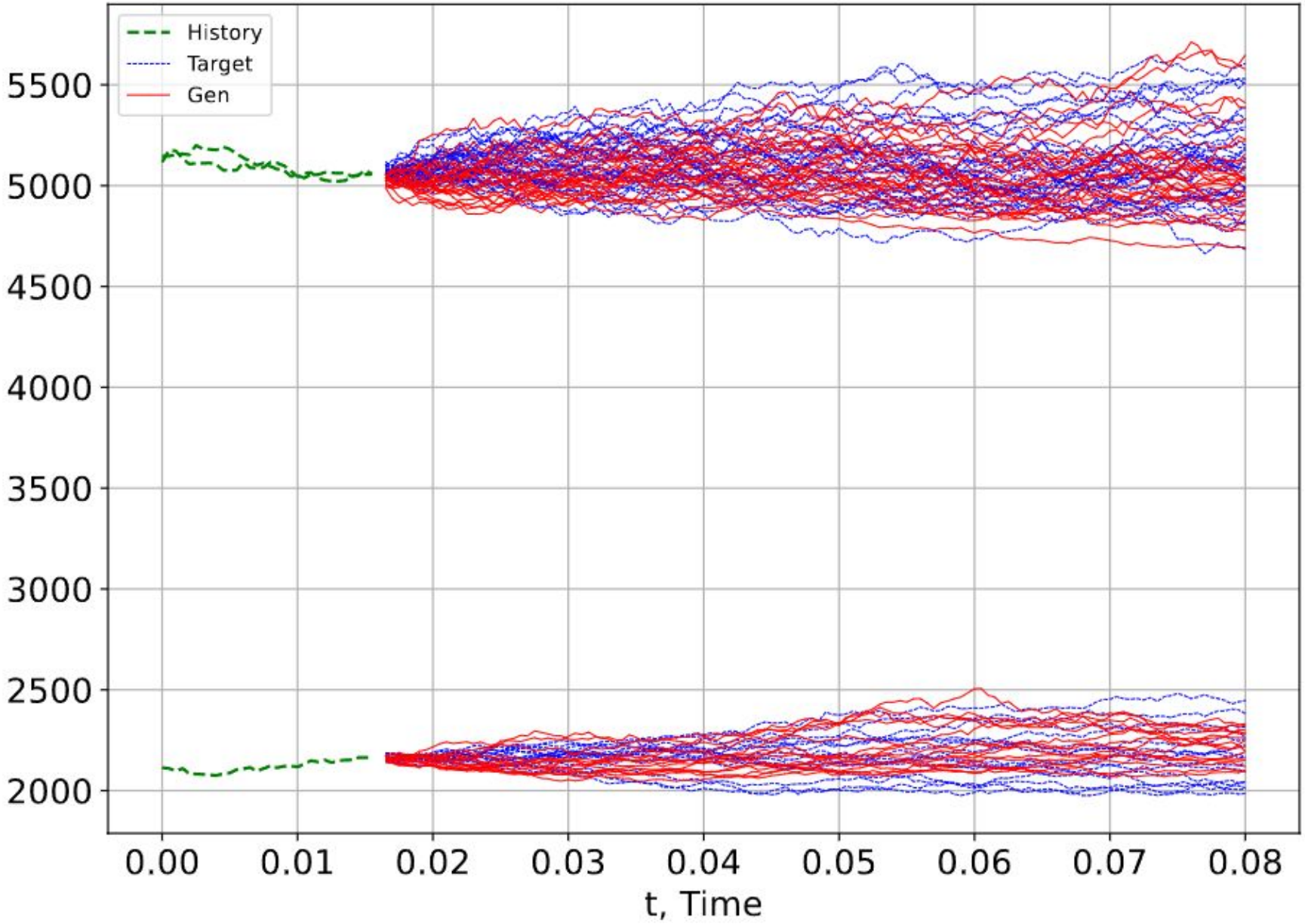
Experiments (GBM 1D)



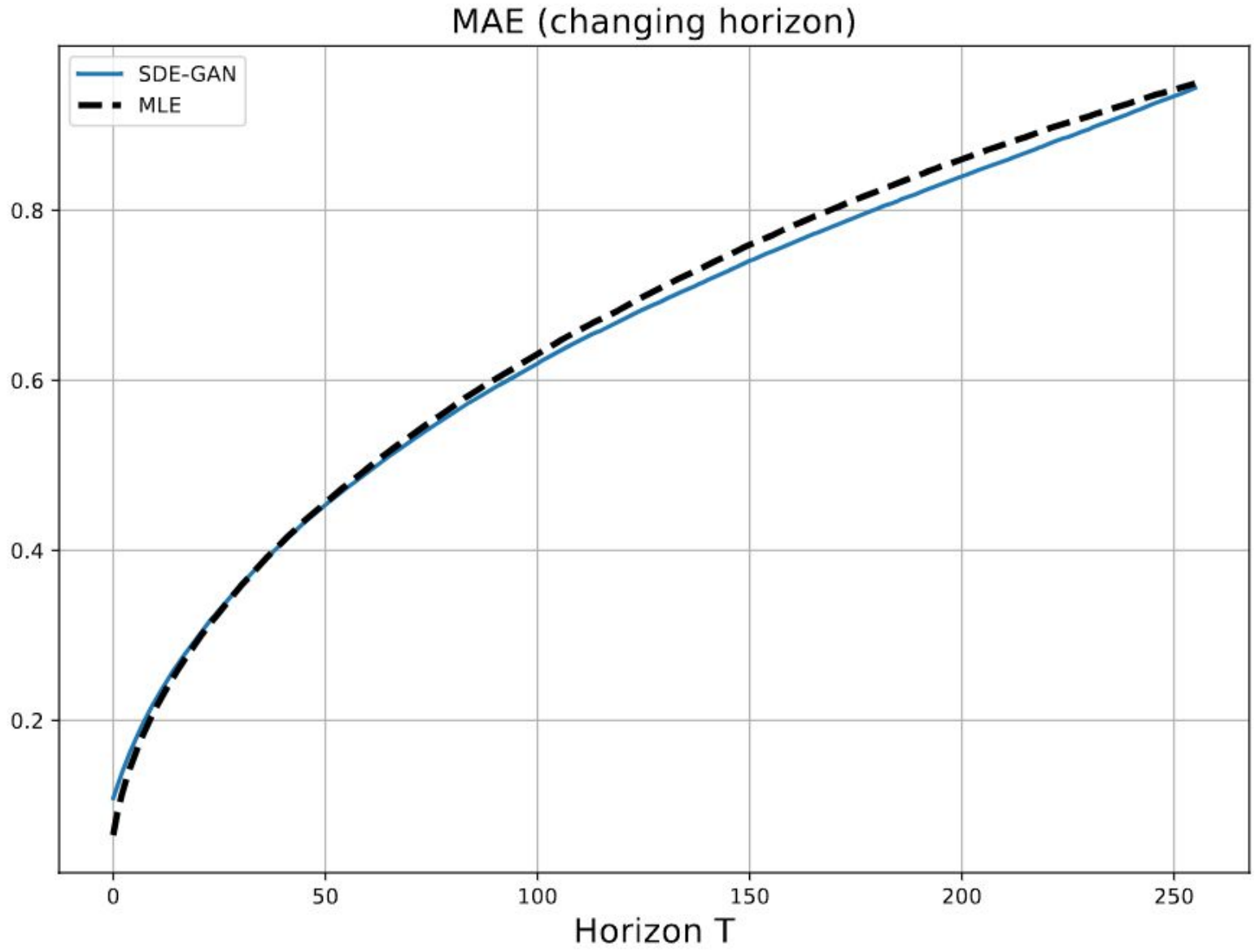
Experiments (GBM 1D)



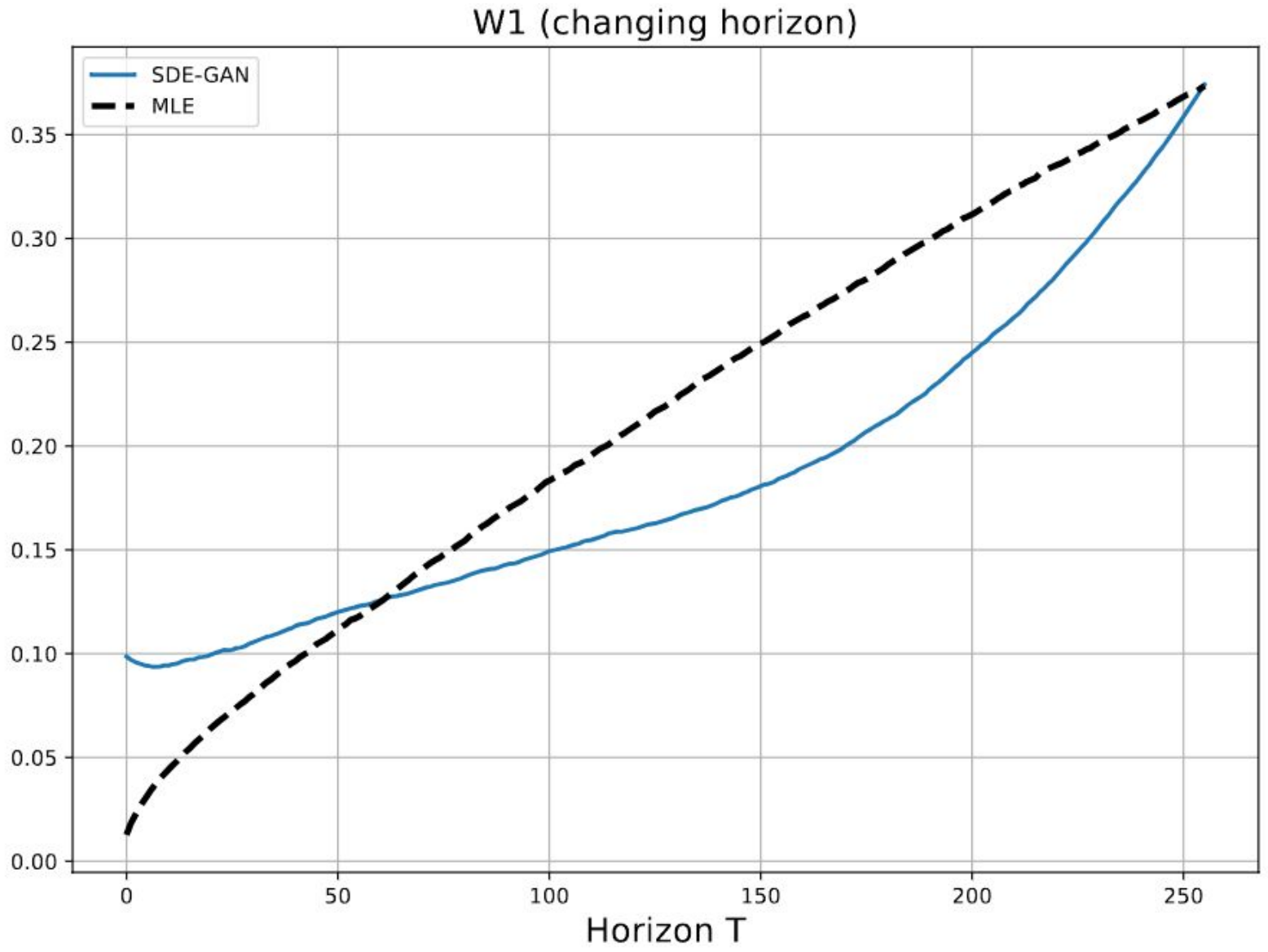
Experiments (GBM 1D, Samples from SDE-GAN)



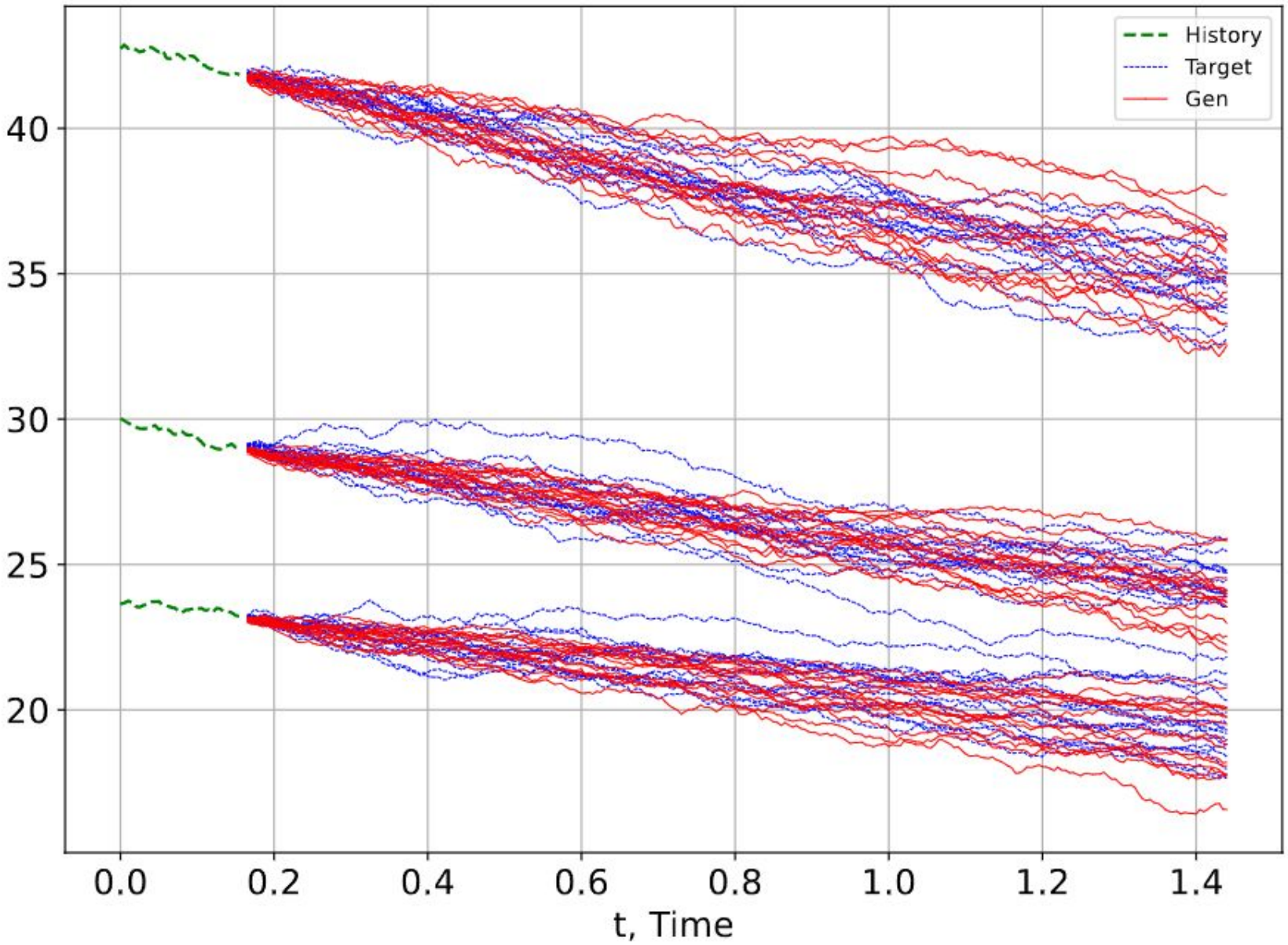
Experiments (CIR 1D)

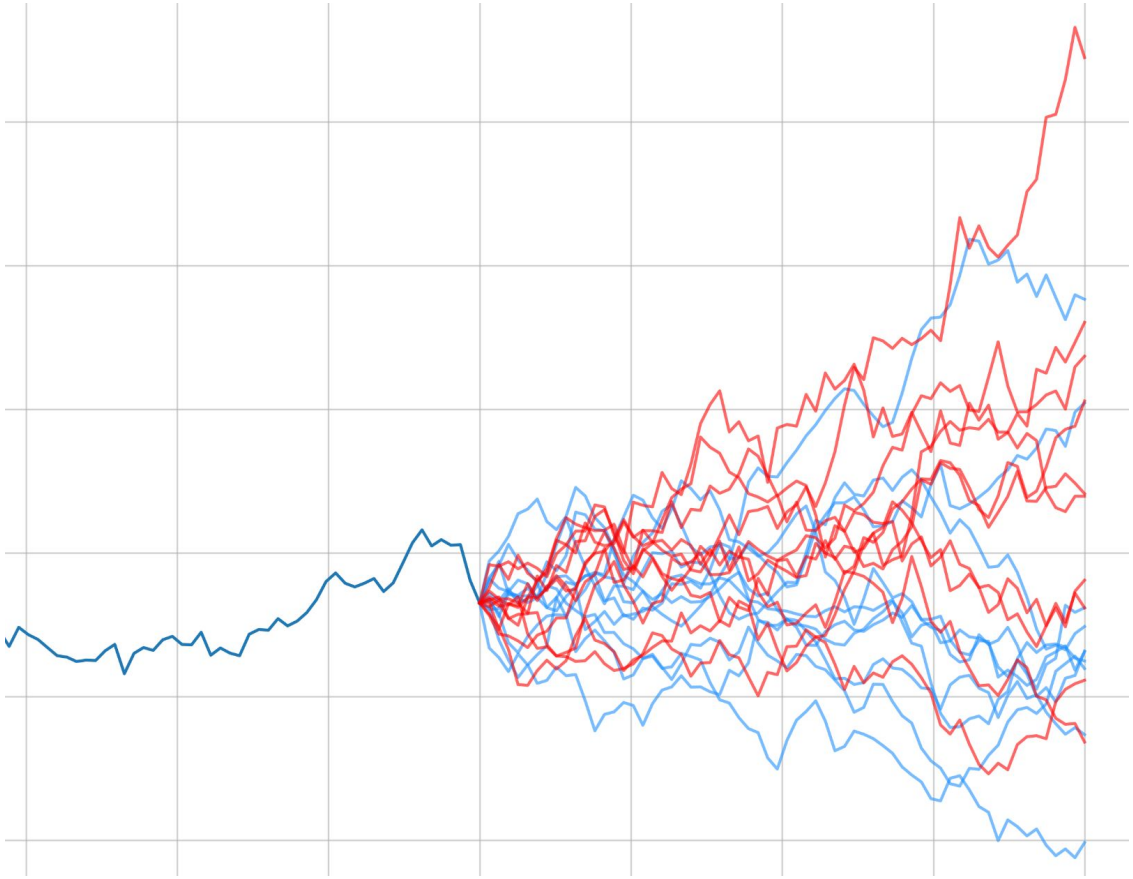


Experiments (CIR 1D)



Experiments (CIR 1D, Samples from SDE GAN)





What's more?

Conclusions

We're still on the way, meanwhile…

Still to explore:

1. **Multivariate** time series applications (finance, weather, sensors…) are under-explored

2. **Jump diffusions** (Jump diffusions and Lévy diffusions)

3. **Downstream tasks** (Model-based optimal control, optimal stopping, filtering…)



That’s the end :)

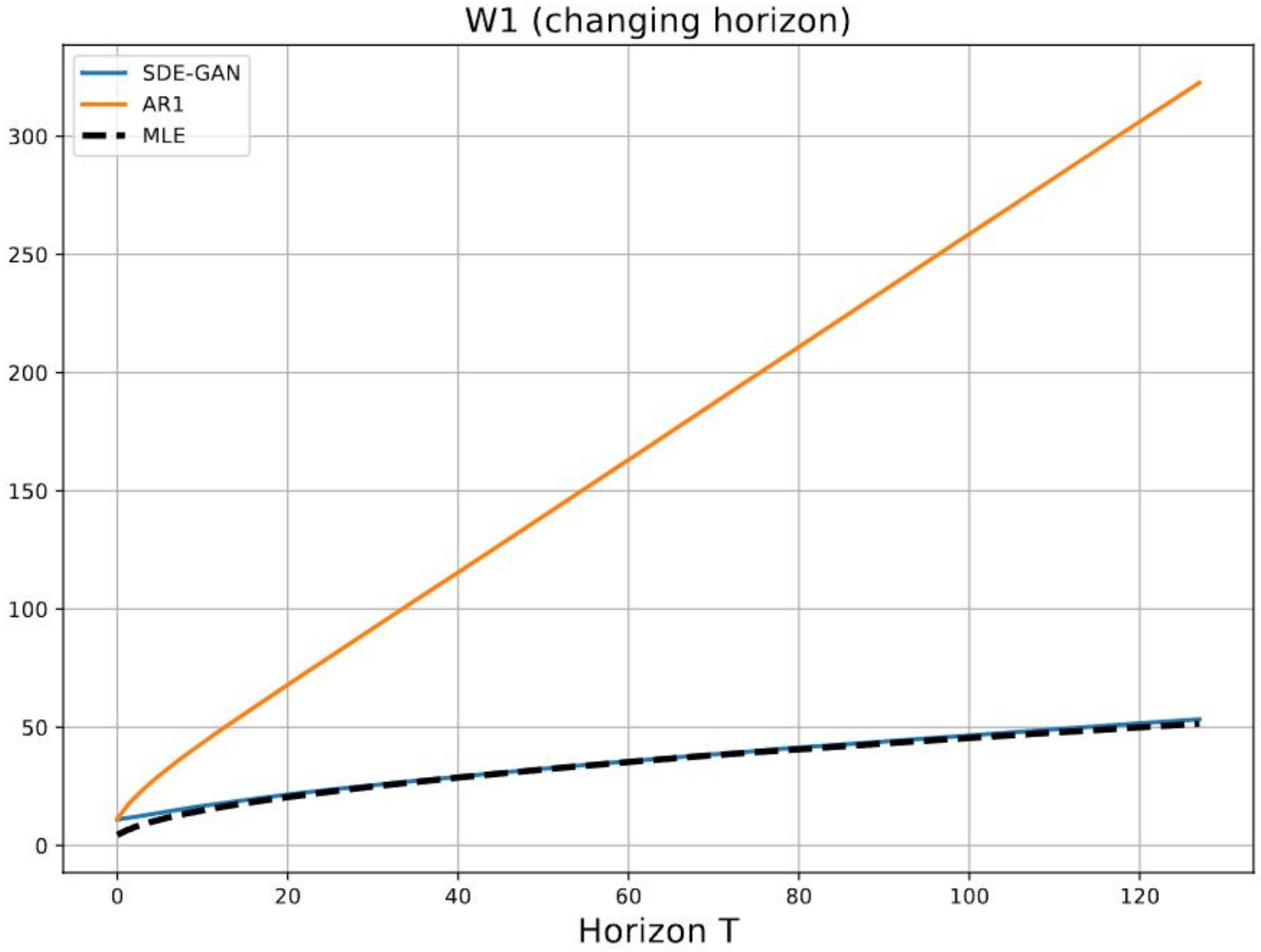
Стохан24



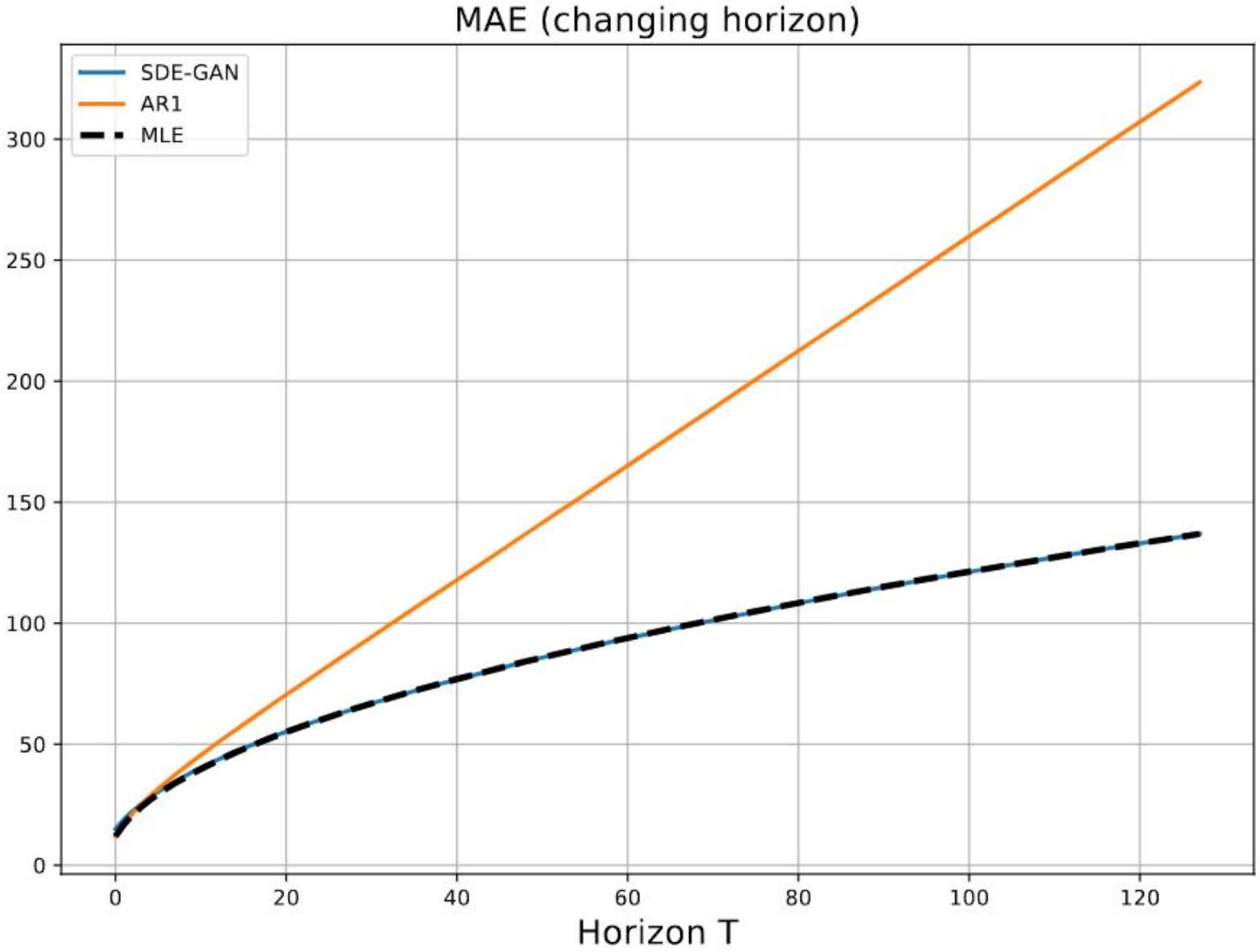
APPENDIX



Full charts of metrics in changing horizon (GBM)



Full charts of metrics in changing horizon (GBM)



Why Data Matters

